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Last login: Wed Jun 24 20:09:03 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % mkdir -p "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly"

find ~/Downloads -name "minimd_dfly_ppn1.py" 2>/dev/null
find ~/Downloads -name "run_minimd_dfly_sweep.sh" 2>/dev/null
/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly/minimd_dfly_ppn1.py
/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly/run_minimd_dfly_sweep.sh
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly"
chmod +x run_minimd_dfly_sweep.sh
export DFLY_SIZE="4"
export STATS_DIR="./results"
sst minimd_dfly_ppn1.py
=====
Topology           : Dragonfly (size 4) - MiniMD
hosts_per_router   : 1
routers_per_group  : 2
num_groups         : 2
Total nodes        : 4
Total MPI ranks    : 4
PPN (ranks/node)   : 1    <-- should be 1
Stats CSV          : ./results/minimd_dragonfly_4_ppn1_nodes4_stats.csv
=====
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=4
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=4
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=4
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=4
161473537282:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=4
161473537282:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=4
AllPingPong: total time 161476.347 us, loop 325, bufLen 20510, latency 248.425 us. bandwidth 0.0
82560 GB/s
161476527798:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=4
161476527798:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=4
317171011766:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=4
317171011766:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=4
AllPingPong: total time 155695.826 us, loop 325, bufLen 9000, latency 239.532 us. bandwidth 0.03
7573 GB/s
317172353865:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=4
317172353865:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=4
478645711449:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=4
478645711449:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=4
AllPingPong: total time 161476.348 us, loop 325, bufLen 20510, latency 248.425 us. bandwidth 0.0
82560 GB/s
478648701965:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=4
478648701965:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=4
AllPingPong: total time 155695.827 us, loop 325, bufLen 9000, latency 239.532 us. bandwidth 0.03
7573 GB/s
Allreduce: ranks 4, loop 40, 1 double(s), latency 2454.565 us
Barrier: ranks 4, loop 15, latency 1395.619 us
Simulation is complete, simulated time: 753.467 ms
stephenrajva@Stephens-MacBook-Air-4 DragonFly % ./run_minimd_dfly_sweep.sh
=== Running MiniMD dragonfly size 4 (PPN=1) ===
=====
Topology           : Dragonfly (size 4) - MiniMD
hosts_per_router   : 1
routers_per_group  : 2
num_groups         : 2
Total nodes        : 4
Total MPI ranks    : 4
PPN (ranks/node)   : 1    <-- should be 1
Stats CSV          : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly/
results/minimd_dragonfly_4_ppn1_nodes4_stats.csv
=====
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=4
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=4
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=4
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=4
161473537282:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=4

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161473537282:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=4
AllPingPong: total time 161476.347 us, loop 325, bufLen 20510, latency 248.425 us. bandwidth 0.0
82560 GB/s
161476527798:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=4
161476527798:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=4
317171011766:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=4
317171011766:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=4
AllPingPong: total time 155695.826 us, loop 325, bufLen 9000, latency 239.532 us. bandwidth 0.03
7573 GB/s
317172353865:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=4
317172353865:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=4
478645711449:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=4
478645711449:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=4
AllPingPong: total time 161476.348 us, loop 325, bufLen 20510, latency 248.425 us. bandwidth 0.0
82560 GB/s
478648701965:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=4
478648701965:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=4
AllPingPong: total time 155695.827 us, loop 325, bufLen 9000, latency 239.532 us. bandwidth 0.03
7573 GB/s
Allreduce: ranks 4, loop 40, 1 double(s), latency 2454.565 us
Barrier: ranks 4, loop 15, latency 1395.619 us
Simulation is complete, simulated time: 753.467 ms
=== Running MiniMD dragonfly size 16 (PPN=1) ===
=====
Topology          : Dragonfly (size 16) - MiniMD
hosts_per_router  : 1
routers_per_group : 4
num_groups        : 4
Total nodes       : 16
Total MPI ranks   : 16
PPN (ranks/node)  : 1  <-- should be 1
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly/
results/minimd_dragonfly_16_ppn1_nodes16_stats.csv
=====
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
161476956324:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
AllPingPong: total time 161479.766 us, loop 325, bufLen 20510, latency 248.430 us. bandwidth 0.0
82558 GB/s
161479946840:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
161612889590:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
161614156223:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
161614156223:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
161615813406:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
161617080039:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
161617080039:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
161737554558:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
161740411674:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
161742786506:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
161743815687:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
161743815687:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
161745643622:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
161746672803:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
161746672803:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
317174430808:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
AllPingPong: total time 155695.826 us, loop 325, bufLen 9000, latency 239.532 us. bandwidth 0.03

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7573 GB/s

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317175772907:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
317353585674:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
317354852307:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
317354852307:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
317354994473:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
317356261106:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
317356261106:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
317521472242:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
317522947741:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
317526704190:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
317527733371:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
317527733371:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
317528179689:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
317529208870:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
317529208870:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
478657028438:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
AllPingPong: total time 161484.246 us, loop 325, bufLen 20510, latency 248.437 us. bandwidth 0.0
82556 GB/s
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```
478660018954:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
478958483757:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
478959750390:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
478959750390:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
478961407573:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
478962674206:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
478962674206:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
479261800673:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
479262829854:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
479262829854:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
479264290584:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
479264657789:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
479265686970:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
479265686970:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
479267147700:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
AllPingPong: total time 155695.827 us, loop 325, bufLen 9000, latency 239.532 us. bandwidth 0.03
7573 GB/s
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Allreduce: ranks 16, loop 40, 1 double(s), latency 2475.783 us

Barrier: ranks 16, loop 15, latency 1399.578 us

Simulation is complete, simulated time: 754.392 ms

=== Running MiniMD dragonfly size 64 (PPN=1) ===

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```
Topology          : Dragonfly (size 64) - MiniMD
hosts_per_router  : 1
routers_per_group : 8
num_groups        : 8
Total nodes       : 64
Total MPI ranks   : 64
PPN (ranks/node)  : 1  <-- should be 1
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly/
results/minimd_dragonfly_64_ppn1_nodes64_stats.csv
=====
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180000:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=64
180000:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=64
180000:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=64
180000:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=64
180000:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=64
180000:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=64
180000:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=64
180000:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=64
180000:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=64
180000:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=64
180000:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=64
180000:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=64
180000:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=64
180000:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=64
180000:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=64
180000:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=64
180000:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
180000:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
180000:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=64
```

180000:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=64
180000:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=64
180000:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=64
180000:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=64
180000:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=64
180000:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=64
180000:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=64
180000:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=64
180000:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=64
180000:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=64
180000:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=64
180000:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=64
180000:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=64
180000:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=64
180000:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=64
180000:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=64
180000:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=64
180000:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=64
180000:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=64
180000:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=64
180000:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=64
180000:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=64
180000:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=64
180000:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=64
180000:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=64
180000:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=64
180000:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=64
180000:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=64
180000:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=64
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=64
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=64
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=64
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=64
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=64
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=64
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=64
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=64
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=64
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=64
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=64
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=64
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=64
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=64
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=64
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=64
161748144517:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=64
161751068333:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=64
161774362286:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=64
161777219402:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=64
161834465656:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=64
AllPingPong: total time 161837.209 us, loop 325, bufLen 20510, latency 248.980 us. bandwidth 0.0
82376 GB/s
161837389418:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=64
161843065954:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=64
161845923043:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=64
161972440609:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=64
161972440609:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=64
161972440609:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=64
161973023567:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=64
161973023567:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=64
161973023567:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=64
161975364243:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=64
161975364243:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=64
161975364243:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=64
161975946945:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=64
161975946945:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=64
161975946945:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=64
162022502294:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=64
162024345215:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=64
162025425571:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=64

162026895156:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=64
162027202331:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=64
162027411414:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=64
162027525471:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=64
162028041729:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=64
162029752218:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=64
162030268092:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=64
162030382533:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=64
162030898407:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=64
162265637134:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=64
162268494250:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=64
162269074185:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=64
162271931301:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=64
162332869400:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=64
162335726516:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=64
162339859560:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=64
162342716676:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=64
162468761312:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=64
162468761312:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=64
162468761312:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=64
162469447655:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=64
162469447655:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
162469447655:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=64
162471618374:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=64
162471618374:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=64
162471618374:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=64
162472304872:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=64
162472304872:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=64
162472304872:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=64
162520546525:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=64
162521062783:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
162523235869:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=64
162523403587:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=64
162523558030:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=64
162523919461:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=64
162523922212:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=64
162524074288:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=64
162526092931:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=64
162526415092:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=64
162526779429:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=64
162526930966:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=64
317488840601:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=64
317490249400:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=64
317558333330:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=64
317559808829:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=64
317575161740:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=64
AllPingPong: total time 155739.181 us, loop 325, bufLen 9000, latency 239.599 us. bandwidth 0.03
7563 GB/s
317576570539:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=64
317626983638:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=64
317628459137:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=64
317713989119:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=64
317713989119:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=64
317713989119:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=64
317715359804:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=64
317715359804:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=64
317715359804:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=64
317715397918:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=64
317715397918:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=64
317715397918:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=64
317716768603:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=64
317716768603:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=64
317716768603:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=64
317763197711:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=64
317764606510:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=64
317808266901:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=64
317809742400:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=64
317811836685:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=64
317813103318:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=64
317813122661:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=64

317813312184:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=64
317814394630:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=64
317814578817:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=64
317814598160:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=64
317815870129:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=64
318049554818:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=64
318051030317:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=64
318052991869:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=64
318054895966:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=64
318116787084:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=64
318118262583:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=64
318123777244:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=64
318125252743:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=64
318252678996:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=64
318252678996:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=64
318252678996:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=64
318253433373:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=64
318253433373:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
318253433373:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=64
318254154495:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=64
318254154495:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=64
318254154495:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=64
318254908872:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=64
318254908872:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=64
318254908872:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=64
318307153553:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=64
318307909520:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=64
318309455310:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=64
318311063319:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=64
318415282102:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=64
318415866010:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
318416928353:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=64
318417512261:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=64
318418293607:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=64
318418877515:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=64
318419939858:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=64
318420523766:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=64
479147544907:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=64
479150468669:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=64
479398316897:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=64
AllPingPong: total time 161824.670 us, loop 325, bufLen 20510, latency 248.961 us. bandwidth 0.0
82382 GB/s
479401240659:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=64
479646894457:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=64
479649751573:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=64
479669633154:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=64
479669633154:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=64
479669633154:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=64
479670216112:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=64
479670216112:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=64
479670216112:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=64
479672556788:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=64
479672556788:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=64
479672556788:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=64
479673139490:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=64
479673139490:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=64
479673139490:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=64
479905416987:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=64
479906274082:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=64
479908274049:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=64
479909131171:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=64
479930740976:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=64
479932583897:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=64
479933664253:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=64
479935441013:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=64
479946122663:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=64
479946638921:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=64
479948979725:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=64
479949495599:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=64
480142810291:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=64

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480143496634:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=64
480145667353:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=64
480146353851:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=64
480149939854:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=64
480150456112:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=64
480152796916:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=64
480153312790:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=64
480394499406:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=64
480397356522:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=64
480646602728:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=64
480647118986:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=64
480649459790:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=64
480649975664:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=64
480653761639:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=64
480653761639:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=64
480653761639:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=64
480654789486:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=64
480654789486:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
480654789486:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=64
480656618701:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=64
480656618701:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=64
480656618701:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=64
480657646447:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=64
480657646447:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=64
480657646447:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=64
480928825102:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=64
480929341360:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
480929569474:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=64
480930768073:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=64
480931682164:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=64
480932198038:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=64
480932426536:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=64
480933624906:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=64
AllPingPong: total time 155739.181 us, loop 325, bufLen 9000, latency 239.599 us. bandwidth 0.03
7563 GB/s
Allreduce: ranks 64, loop 40, 1 double(s), latency 2504.418 us
Barrier: ranks 64, loop 15, latency 1403.452 us
Simulation is complete, simulated time: 756.386 ms
All MiniMD dragonfly runs finished. Results in: /Users/stephenrajva/Downloads/HPC-Upload 2/SST_S
timulator/MiniMD/DragonFly/results
stephenrajva@Stephens-MacBook-Air-4 DragonFly %
stephenrajva@Stephens-MacBook-Air-4 DragonFly % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/S
ST_Stimulator/MiniMD/DragonFly"

cp "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/ppn1/analyze_universal
.py" .

python3 - << 'EOF'
import re
f = "analyze_universal.py"
s = open(f).read()
s = re.sub(r'STUDY\s*=\s*".*?"', 'STUDY = "MiniMD_Dragonfly"', s, count=1)
s = re.sub(r'TOPOLOGY\s*=\s*".*?"', 'TOPOLOGY = "other"', s, count=1)
new_runs = '''RUNS = [
    ("4", (0,), "minimd_dragonfly_4_ppn1_nodes4_stats.csv"),
    ("16", (0,), "minimd_dragonfly_16_ppn1_nodes16_stats.csv"),
    ("64", (0,), "minimd_dragonfly_64_ppn1_nodes64_stats.csv"),
]'''
s = re.sub(r'RUNS = \[.*?\]', new_runs, s, count=1, flags=re.DOTALL)
open(f, "w").write(s)
print("Updated for MiniMD dragonfly.")
EOF

export DATA_DIR="./results"
python3 analyze_universal.py
Updated for MiniMD dragonfly.

MiniMD_Dragonfly
Size      Latency(ns)  Hops      Thru(GB/s)  Packets    Bits          Idle%    Stalls
-----

```

4	158.22	-	0.2046	99472	1233338240	87.4	0
16	252.13	-	1.3795	672332	8325396640	92.2	0
64	511.70	-	6.0640	2964220	36693812000	95.3	0

Saved: MiniMD_Dragonfly_latency_ns.png

(skipping Average hop count: not available for topology 'other')

Saved: MiniMD_Dragonfly_throughput.png

Saved: MiniMD_Dragonfly_packets.png

Saved: MiniMD_Dragonfly_bits.png

Saved: MiniMD_Dragonfly_idle_pct.png

Saved: MiniMD_Dragonfly_stalls.png

stephenrajva@Stephens-MacBook-Air-4 DragonFly %